



# Closed-Loop Control Systems Robust to Objective and Environment Shift

Learning-based network resource management and other closed-loop control that adapt through policy transfer and LLM agents, implemented and validated through Sim2Real

- ① **Policy Transfer** — Network Resource Management & Cloud Robotics
- ② **Agentic AI** — LLM Agents for System Operation
- ③ **Sim2Real** — Purpose-Built Simulators, Surrogates, Testbeds & Deployment

## Youngseok Lee

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Advisor: Prof. Honguk Woo (CSI-Agent Group) · Expected graduation Feb. 2027

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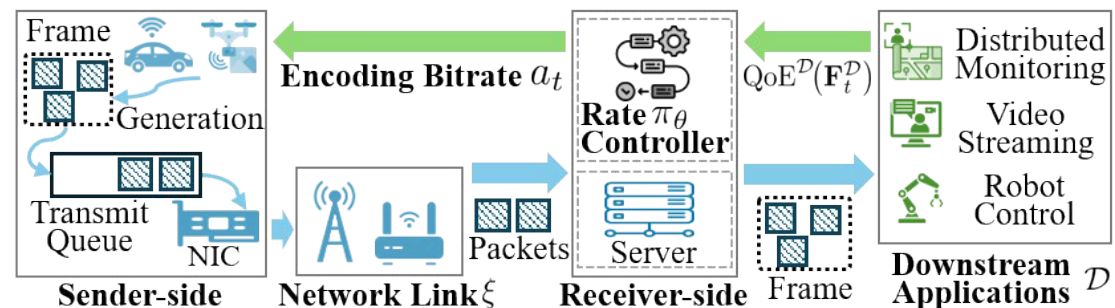
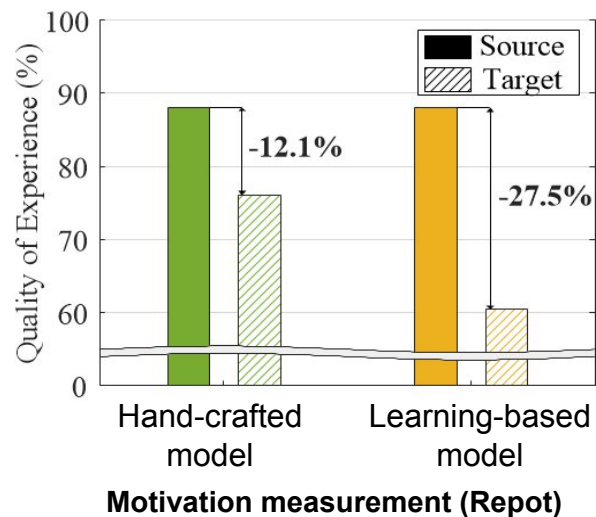
# Youngseok Lee — Ph.D. Candidate, Sungkyunkwan Univ.

- **Education** — Integrated M.S.–Ph.D., ECE, Sungkyunkwan University (Mar. 2013 – Feb. 2027, expected) · Advisor: Prof. Honguk Woo (CSI-Agent Group) · Dissertation (tentative): Transferable Reinforcement Learning for Quality-Driven Network Control
  - 2018: moved to Prof. Honguk Woo's group and settled on the dissertation topic · Expert Research Personnel service (Mar. 2017 – Mar. 2020) · Global Ph.D. Fellowship, 5-year term (2013–2018, Ministry of Education / NRF Korea)
- **Achievements** — 8 granted patents (US 5 / KR 3, first inventor on 4) · 5 systems delivered · 3 SCIE journal papers · 1 NeurIPS 2024 paper · 1 IEEE INFOCOM 2027 submission under review · 35 earlier publications (2012–2018)
  - Delivered to Samsung Electronics ×2, Polaris AI, Acryl, and ADD (Agency for Defense Development), each as technical lead · 20 funded projects (principal investigator 1 · technical lead 10 · researcher 9) · 9 software copyrights
- **Research coverage** — Network resource management and other closed-loop control robust to objective and environment shift, achieved through policy transfer and LLM agents that automate system operation, and validated through Sim2Real

| Research area                                                            | Representative work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Outcomes                                                                                                                                                                                        |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ① Policy Transfer — Network Resource Management & Cloud Robotics         | <ul style="list-style-type: none"> <li>○ XferQoE — One QoS-pretrained rate controller, adapted per application by objective-weight fitting</li> <li>○ Repot — Bandwidth allocation transferred across network environments by flow embedding and action shaping</li> <li>○ Cloud-based multi-robot collaborative control with action-downlink support (XferQoE follow-up, in progress)</li> </ul>                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>○ IEEE INFOCOM 2027 (first author, under review)</li> <li>○ IEEE Access 2021 (first author)</li> <li>○ 1 first-inventor patent family (KR-US)</li> </ul> |
| ② Agentic AI — LLM Agents for System Operation                           | <ul style="list-style-type: none"> <li>○ AgentV2G (in progress) — LLM-agent V2G fleet operation under drift, day-ahead → real-time as one pipeline</li> <li>○ LDuS — LLM-guided skill diffusion for zero-shot policy adaptation</li> </ul>                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>○ NeurIPS 2024 (second author)</li> </ul>                                                                                                                |
| ③ Sim2Real — Purpose-Built Simulators, Surrogates, Testbeds & Deployment | <ul style="list-style-type: none"> <li>○ 3 application evaluation environments (robot · monitoring · streaming); JAX/Flax training framework</li> <li>○ V2G fleet-operation environment — Market rules, system marginal price (SMP) series, and EV charging-session records</li> <li>○ Packet-level C++ training simulator (S-Sim) → NS-3 · Raspberry Pi NiL testbed</li> <li>○ NVMe-oF storage simulator with dual-output DNN surrogate</li> <li>○ QLED wireless streaming testbed; edge-server inference pipeline</li> <li>○ LTE band-selection service; GPS P(Y) acquisition analysis tool</li> </ul> | <ul style="list-style-type: none"> <li>○ Delivered ×5 (Samsung ×2, Polaris AI, Acryl, ADD)</li> <li>○ 2 US patents with Samsung (storage)</li> </ul>                                            |

# Control Robust to Objective and Environment Shift

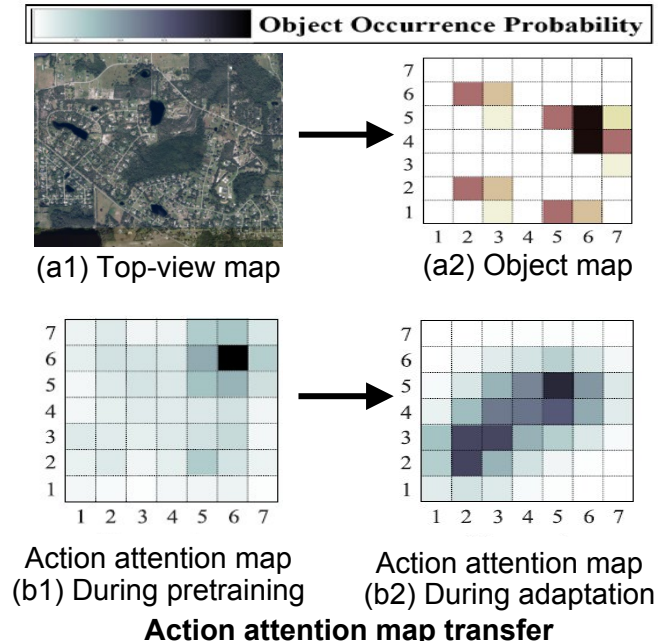
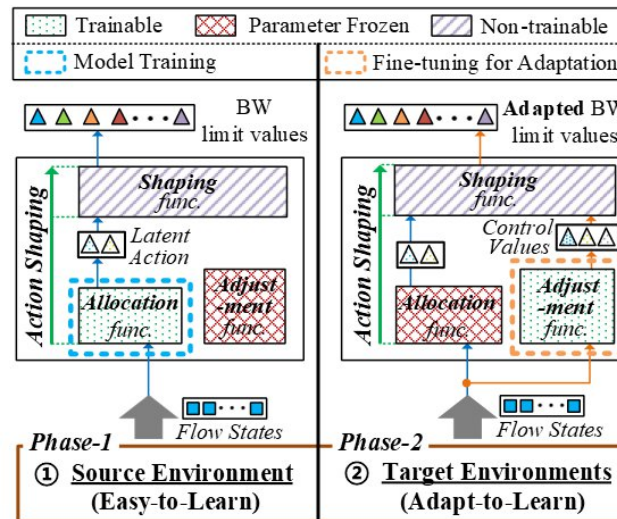
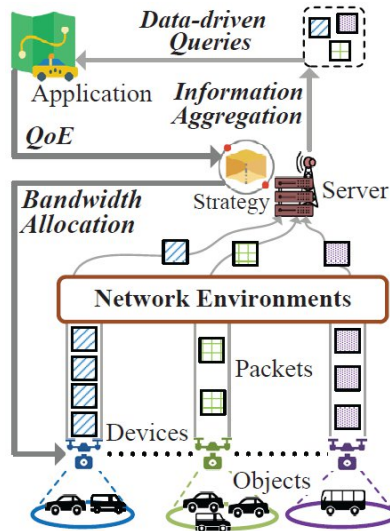
- **Problem** — A learned controller outperforms a hand-crafted model in the source environment, yet degrades far more severely than the hand-crafted model when the objective or the environment shifts even slightly between source and target (left figure)
  - Such shifts occur after deployment as changes in the application objective (the definition of QoE, the quality of experience), network condition, demand pattern, or market rule; the hand-crafted model degrades gracefully but requires manual re-tuning for each condition, whereas the learned model needs data collection and retraining for each condition
- **Goal** — Closed-loop control that combines the robustness of the hand-crafted model with the performance of the learned model and adapts without human intervention
  - ① **Policy Transfer** — A single trained rate-control or bandwidth-allocation policy is transferred to a new objective or network environment without retraining, or with a small amount of data (XferQoE · Repot)
  - ② **Agentic AI** — LLM agents translate rules and requirements into executable objectives and code, and validate the outcome through a feedback loop, automating system operation (AgentV2G · LDuS)
  - ③ **Sim2Real** — The two approaches are implemented and validated on purpose-built simulators, surrogate models, testbeds, and fielded systems; five such systems were delivered (Samsung ×2, Polaris AI, Acryl, and ADD)
- **Representative system** — Network-control closed loop (right figure): rate controller → network link → application QoE feedback



Closed-loop QoE-driven adaptive rate control (XferQoE)

# Report — Transferable RL for Networked Monitoring

- **Overview** — 2020–2021 · IEEE Access 2021 (first author) · Network Monitoring System patent, KR 2023 / US 2026 (first inventor)
- **Problem** — When sensing devices share a single link, the bandwidth allocation must keep a server's continual query accurate as the network degrades; re-learning that allocation from scratch for each new network scale and condition is prohibitively expensive (left figure)
- **Method** — Two-phase model training and adaptation (center figure): ① The RL (reinforcement learning) policy is trained in an ideal time-slotted source environment where samples are inexpensive → ② Only its adjustment function is fine-tuned in an NS-3 target environment where sample collection is expensive
  - **Flow embedding** — Each device's stream is represented by its own history and its relation to the other streams in a common vector space, so that a single policy transfers across network scales and conditions
  - **Action shaping** — The frozen allocation function produces a latent action that a non-trainable shaping function maps to bandwidth limits; a small adjustment function sets the shaping function's control values and is the only component fine-tuned in the target
- **Result** — The policy adapts with 6.25% of the samples required for training from scratch and reaches 14.5–20.8% higher QoE than heuristic and learning-based baselines in the target environment; QoE relative to the equal-share reference remains at 91–94% across 49–400 devices in the source environment; the SUMO (Simulation of Urban MObility) case study yields +6.5% (urban) and +12.6% (suburban) over Top-Opt, a Top-K heuristic (right figure)



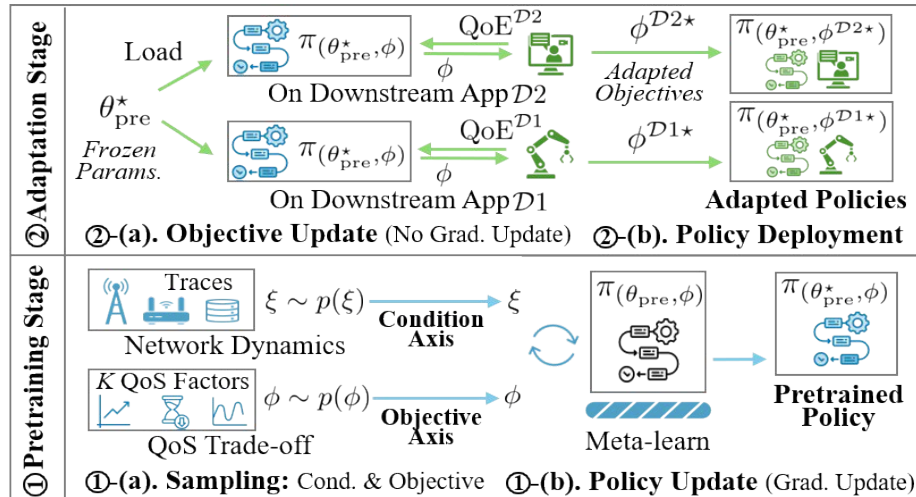
Networked monitoring systems

Action shaping model structure

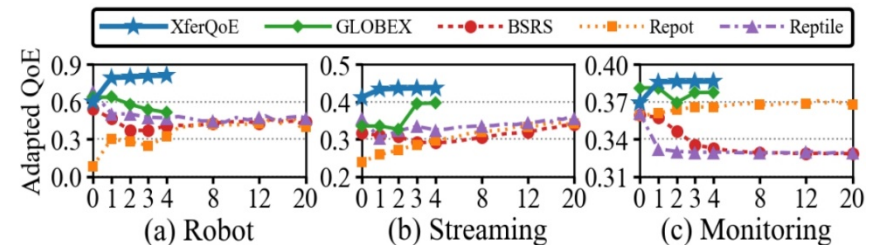
Action attention map transfer

# XferQoE — QoS-to-QoE Transfer for Adaptive Rate Control

- **Overview** — 2024–2026 · IEEE INFOCOM 2027 (first author, under review) · Physical-AI Core-Technology PoC, MSIT (Ministry of Science and ICT) / NIPA (National IT Industry Promotion Agency), 2025
- **Problem** — Each application weighs the QoS (quality of service) factors (timeliness, throughput, smoothness) in its own way, and its QoE (quality of experience) is observable only through the complete network-to-application stack, so training a controller per application pays both costs again: the per-application objective design and the full-stack QoE collection
- **Method** — ① A single rate controller is meta-pretrained on QoS objectives alone in S-Sim (Simplified Simulator), with training tasks sampled across bandwidth dynamics and objective weights → ② The policy is then frozen, and only the objective weight  $w$  is fitted from a few QoE feedback shots — no gradient update (left figure)
  - A regime–context encoder infers the bandwidth-dynamics regime and the step context (condition axis)
  - Factor critics compose the value from one head per QoS factor under  $w$  (objective axis)
  - Policy inference is executed on the edge server, and only observations traverse the link
- **Result** — XferQoE outperforms each application's strongest learning-based baseline by 2–13% on average (up to 24%) within four shots and attains 92% of the E2E (end-to-end) Oracle (a per-application QoE-trained controller); pretrained once in S-Sim, the frozen policy is adapted on NS-3 and on the Raspberry Pi 4 NiL (network-in-the-loop) testbed by weight fitting alone, retaining its advantage in S-Sim, NS-3, and the testbed alike (right figure and table)



Transfer framework: QoS pretraining → QoE adaptation



Adapted QoE per adaptation shot (one shot = one adaptation round)

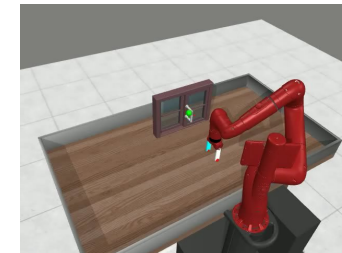
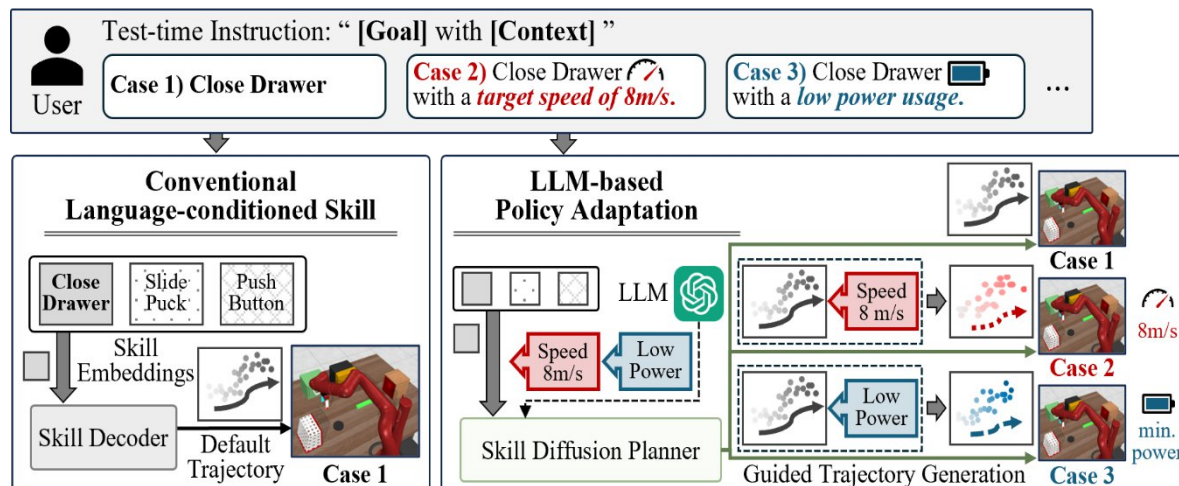
| Method     | QoE    | Success Rate (%) | Frame Quality (%) | Control Continuity (%) | Efficiency (%) | Working Time (s) | Completion Time (s) |
|------------|--------|------------------|-------------------|------------------------|----------------|------------------|---------------------|
| XferQoE    | 0.486  | 70.3             | 22.8              | 90.9                   | 79.8           | 7.7              | 8.72                |
| STEER      | -0.047 | 51.3             | 29.7              | 57.3                   | 83.2           | 7.5              | 15.48               |
| GLOBEX     | 0.104  | 52.8             | 27.0              | 76.2                   | 82.1           | 7.5              | 10.79               |
| Per-app HC | 0.434  | 68.5             | 25.3              | 86.6                   | 82.4           | 7.6              | 9.25                |

NiL testbed, robot control (ManiSkill3 PokeCube)

Per-app HC = per-application hand-crafted controller

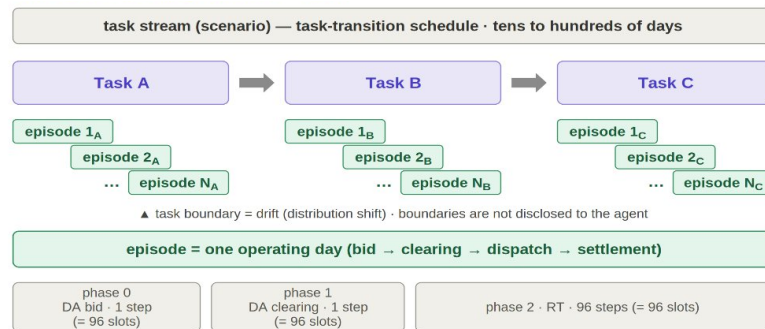
# LDuS — LLM-Guided Skill Diffusion for Policy Adaptation

- **Overview** — 2023–2024 · “LLM-based Skill Diffusion for Zero-shot Policy Adaptation,” NeurIPS 2024 Main Conference Track (second author) · role: algorithm, architecture, and experiment design; robot-task validation
- **Problem** — A skill-based policy learned from expert data cannot adapt to a language-specified context unseen in training, and retraining for each context is expensive (left figure, Cases 2–3). Such a context is a changed environmental condition or a user requirement such as a target speed, low power usage, or a spatial constraint.
- **Method** — An LLM generates a loss function as executable code from the language context, and that loss guides the trajectory generation of a pretrained skill diffusion model, yielding zero-shot policy adaptation
  - **Sequential in-painting** — The observed past state–action sequence is fixed into the trajectory being generated, so the model completes the remaining portion at test time without policy retraining
  - **Iterative closed-loop refinement** — The LLM validates the generated code and the guided trajectory, regenerating faulty code and raising the guidance frequency until the trajectory satisfies the context
- **Result** — LDuS adapts zero-shot to user constraints on how a task is done (speed, force, energy use, spatial bounds) that were never seen in training, with no policy update. Success rate / context reward vs. Diffuser+Guidance (same gradient guidance, hand-written loss): speed constraints on Meta-World (10 goals): 94.6% / 87.4 vs. 76.4% / 69.8; on multi-stage Meta-World (3 long-horizon goals): 84.0% / 82.0 vs. 36.7% / 45.9; hardest case, a force constraint active only while touching the object, contact judged from the camera frame (2 goals, right figure): 93.1% / 63.5 vs. 1.4% / 33.0 · GPT-3.5 turns each constraint into loss-function code; GPT-4 judges contact from the frame, gating when that loss applies

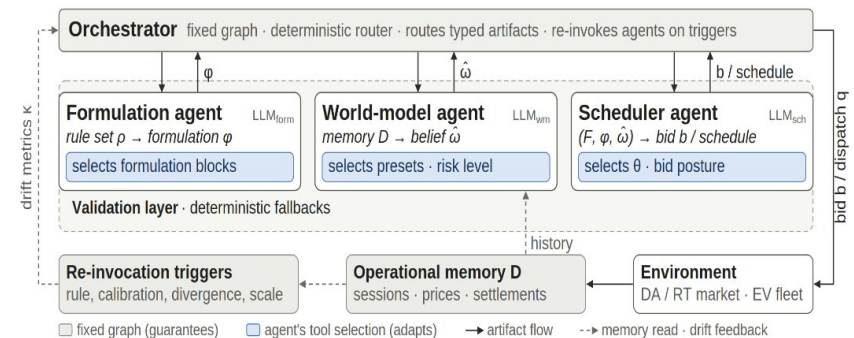


# AgentV2G — Agentic V2G Fleet Scheduling Under Drift

- **Overview** — 2026 – present · Hyundai Motor · Kia V2X (vehicle-to-everything) project (Mar. 2026 – Mar. 2027, technical lead) · paper in preparation (first author)
- **Problem** — A V2G (vehicle-to-grid) charging-hub operator commits irreversible market bids before prices, cleared fractions, and vehicle behavior are revealed, all of which drift over weeks of operation; an agent free to re-plan cannot be trusted with the commitment, while a fixed pipeline cannot follow the drift
- **Method** — A static orchestration graph with dynamic tool selection (right figure): Every operating day executes the same bid → clear → dispatch sequence, so the graph is fixed once, and each agent adapts only by re-selecting among verified tools where drift affects the estimates or rules
  - **Three role agents** — A formulation agent derives the objective and constraints from the market rules and re-derives them when the rules change; a world-model agent maintains a belief state over the unobserved inputs (prices, clearing, sessions) from an operational memory and selects the risk level of its belief corrections; a scheduler agent produces every bid and schedule through an optimization engine (MILP, mixed-integer linear programming)
  - The orchestrator re-invokes an agent when a deterministic drift metric fires and validates every emitted artifact (formulation, belief, schedule) through membership, bounds, and scale checks, falling back to deterministic defaults when a check fails; the objective is immutable within an episode, so no agent can soften the objective it is evaluated against
- **Evaluation** — Day-ahead and real-time market scenarios on real SMP (system marginal price) and EV (electric vehicle) charging-session data, under Jeju pilot rules with EU and NEM (Australian National Electricity Market) presets; fleets of 1–1,000 EVs; a single open-weight LLM (Llama-3.1-8B-Instruct) without fine-tuning. Metrics: episode-level operating quality and recovery under price, fleet-behavior, and market-rule drift, individually and combined (left figure)
  - **Scalability** — A 10,000-EV day is scheduled in ≈54 s, within Hyundai Motor's 90 s requirement (feedback loop disabled)



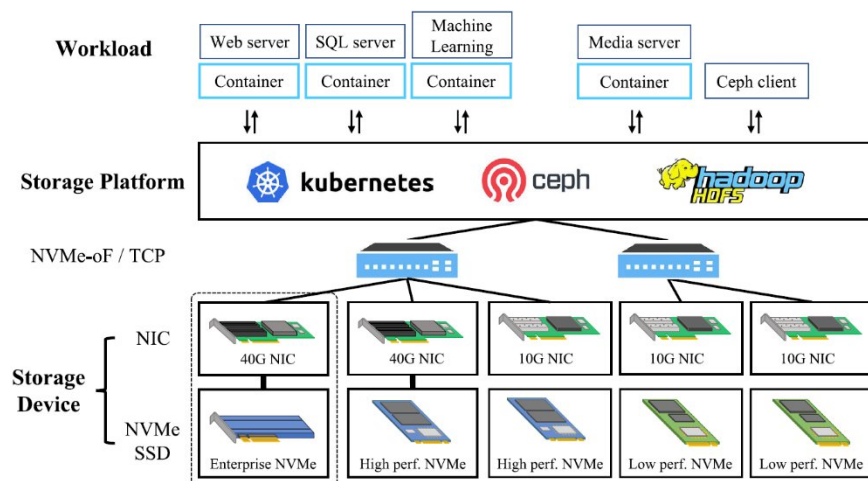
Task stream and operating-day episode structure



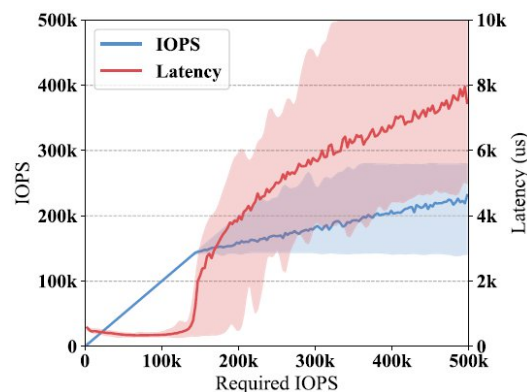
AgentV2G architecture: three role agents on a fixed graph

# NVMe-oF Storage Simulator with a DNN Surrogate

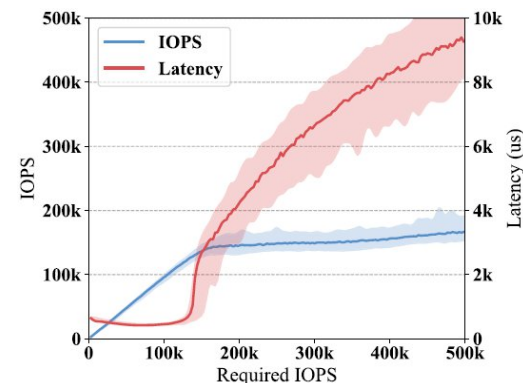
- **Overview** — 2019–2021 · Samsung Electronics project (Aug. 2019 – Jul. 2022, technical lead) · delivered to Samsung Electronics · patents with Samsung: US 12,650,769 B2 (simulation method), US 12,210,760 B2 (migration control) · SW copyright C-2020-037999
- **Purpose** — To reproduce a disaggregated NVMe-oF (NVMe over Fabrics; here NVMe/TCP) cluster in software (NS-3-based) so that storage-management policies (RL policies included) can be tuned for mixed configurations of heterogeneous devices in a data center
- **Surrogate** — FIO (flexible I/O tester) measures per-node serviced IOPS (I/O operations per second) and latency against required IOPS; a single DNN (deep neural network) surrogate spanning device specifications reproduces both responses (right figure: (a) measured pattern vs. (b) pattern estimated by the trained model)
  - Four configurations were profiled, from an RC500 SSD on 10 GbE to a Samsung PM1725a SSD on a 40 GbE NIC (network interface card); the predicted IOPS and latency responses are consistent with the FIO measurements
- **Configurable environment** — Devices are organized under a platform that aggregates them in a two-level hierarchy, so that the environment is reconfigured from a system specification rather than rebuilt (left figure)
  - A tier-level object-migration policy is evaluated across homogeneous and heterogeneous compositions of the measured devices under FIO-generated workloads (the patent-covered scope)



NVMe-oF disaggregated storage systems



(a) Measured performance pattern

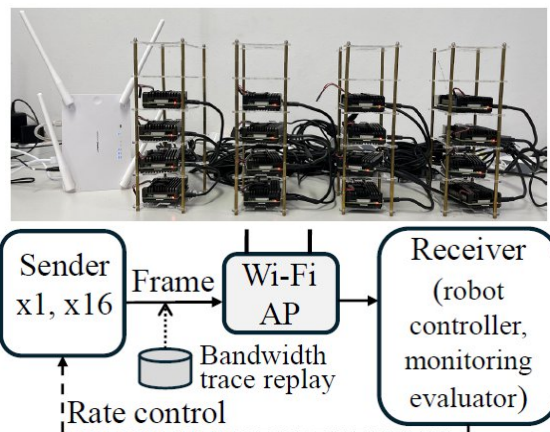


(b) Estimated performance pattern

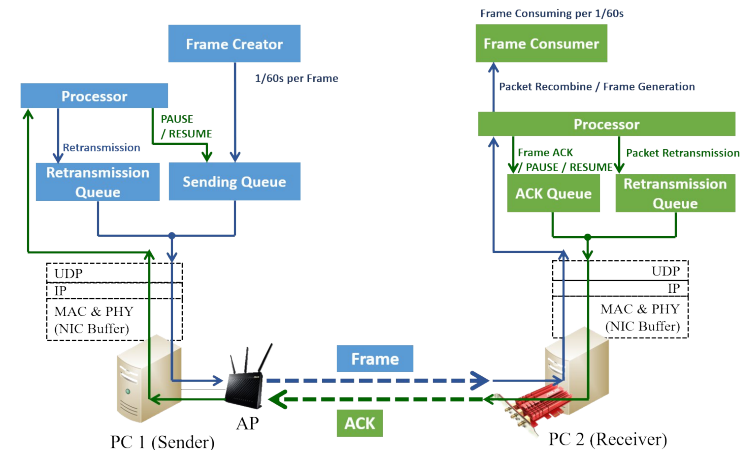
A surrogate model example for NVMe-oF storage

## Real Wireless Links in the Control Loop

- **Raspberry Pi network-in-the-loop (NiL) testbed** (2026; validation path of XferQoE) — Up to 16 Raspberry Pi 4 nodes share a single IEEE 802.11ac link through an ipTIME AX3004R access point to the edge server that runs the application model and the rate controller (left figure)
  - C/C++ TCP/UDP daemons implement episode-level clock synchronization with drift measurement, sender-side virtual qdisc (queueing discipline) queue modeling, and frame-deadline expiry accounting; one wire format and one shared-memory interface span S-Sim, NS-3, and the testbed, so that a single policy runs on all three unchanged
  - **Result** — Policy inference and rate computation require 8.3 ms of each 1 s control interval (STEER baseline: 206.2 ms), so off-board control (policy inference on the edge server rather than on the device) operates with a substantial timing margin
- **Low-latency wireless streaming testbed reproducing the Samsung QLED TV One Connect system** (2018; Samsung Electronics project, Sep. 2017 – Sep. 2018, technical lead · delivered to Samsung Electronics) — Transmitter and receiver Linux hosts use Broadcom BCM4360 3×3 MIMO (multiple-input multiple-output) wireless NICs over 802.11ac (5 GHz) through a legacy ASUS access point; the setup is used for layer-by-layer delay analysis of the standard protocol stack for 4K 60 Hz video (right figure)
  - Three C/C++ components: a TCP sender–receiver with buffer-based flow control, a UDP sender–receiver with bitrate-paced transmission, and a raw-socket sniffer that measures RTT (round-trip time), throughput, packet loss, duplication, and reordering
  - **Result** — Bitrate-paced UDP transmission sustains 4K 60 Hz frames (400 KB per frame, ≈192 Mbps) at 6 ms average RTT and 200 Mbps throughput over 5-minute runs (TCP prototype: 11 ms)



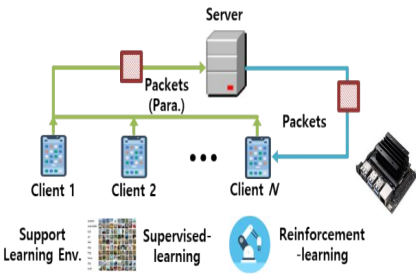
16-node Raspberry Pi 802.11ac NiL testbed



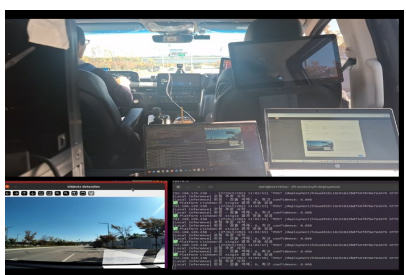
One Connect wireless streaming testbed

# Federated Edge Inference · PS-LTE Network Selection

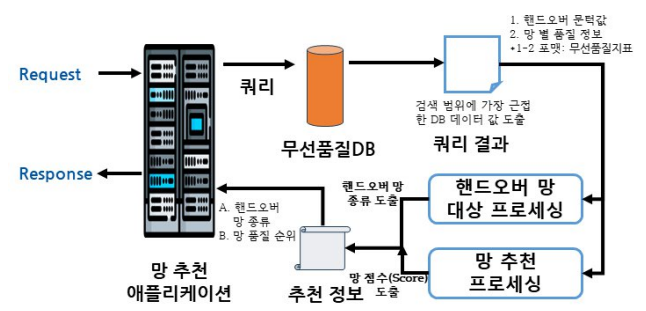
- **Federated inference and learning framework across data-center and edge NPUs (neural processing units)** — MSIT / IITP (Institute for Information & Communications Technology Planning & Evaluation), SW Computing Industry R&D (Apr. 2021 – Jun. 2025) · technical lead · two deliverables
  - **FL (federated learning) testbed built on the open-source Flower framework** (left figure (a)) — Flower (Beutel et al., 2020) was selected over TFF (Google), FATE (WeBank), PFL (Baidu), and PySyft (OpenMined) because it supports PyTorch, FL strategies, networked communication, and embedded deployment; the client side was extended with federated RL and non-IID (non-independent, non-identically distributed) data composition, and the round-based scheduler was improved
    - On a 4-client gRPC testbed with a Jetson Nano (4 GB) client, federated training attains 98% accuracy on MNIST, equal to local training (5-round training time: 5.19 s on the PC clients → 33.48 s on the Jetson)
  - **Edge–server collaborative inference pipeline for real-time traffic sign recognition** (2025; left figure (b): in-vehicle field test at Hwaseong, Gyeonggi) — Delivered to Acryl Inc. (the consortium's industry partner) · role: algorithm design; data provided by KETI (Korea Electronics Technology Institute); models deployed on Acryl's Jonathan platform server and the KETI NPU server
    - A YOLOv8n on the FPGA-based NPU (field-programmable gate array; Xilinx Alveo U250/U280) processes routine frames in real time; when a frame's confidence falls below 0.3 (e.g., at night or in rain), the next frame is forwarded to the server's YOLOv8l for federated inference, improving the detection of damaged or occluded signs on difficult frames
- **Database-backed network selection application for PS-LTE (public-safety LTE) terminals** (Polaris AI, Aug. – Dec. 2024, technical lead · delivered to Polaris AI; right figure) — The application recommends the network a terminal should access, PS-LTE or a commercial LTE band, in areas where the disaster-response and railway networks share a band and interfere
  - A PostgreSQL/PostGIS database of over 200,000 normalized field measurements backs the ranking of 6 LTE bands (PS-LTE B28, commercial B1/B3/B5/B7/B8; 10 carriers) by RSRP, RSRQ, and SINR, adjusted in sparsely covered areas; a Flask server provides the ranking and the threshold-based PS-LTE–commercial handover to a Flutter web application via an HTTP API



(a) Embedded testbed



(b) Deployment



PS-LTE network recommendation system (delivered to Polaris AI)

NPU project: FL testbed and in-vehicle field test



# Summary and Contact

- **Contact** — Youngseok Lee · [yslee.gs@gmail.com](mailto:yslee.gs@gmail.com) · [Google Scholar](#)
- ① **Policy Transfer** — A single trained rate controller or bandwidth-allocation policy is transferred to a new objective (application QoE) or a new environment (network) without retraining, or with a small amount of data (XferQoE · Repot)
- ② **Agentic AI** — LLM agents translate rules and requirements into executable objectives and code, and validate the outcome through a feedback loop, automating system operation (AgentV2G · LDuS)
- ③ **Sim2Real** — Both approaches are implemented and validated on purpose-built simulators, surrogate models, testbeds, and fielded systems; five such systems were delivered (Samsung ×2, Polaris AI, Acryl, and ADD)

## Selected publications and patents (links)

| Type                    | Title                                                                                                       | Venue · Registration                                     | Link                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------|
| Paper · top-tier        | XferQoE: QoS-to-QoE transfer for adaptive rate control                                                      | IEEE INFOCOM 2027 · first author<br>under review         | —                                       |
| Paper · top-tier        | LLM-based skill diffusion for zero-shot policy adaptation (LDuS)                                            | NeurIPS 2024 Main Conference Track ·<br>second author    | <a href="#">NeurIPS</a>                 |
| Paper · SCIE            | Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments | IEEE Access, vol. 9, 2021 · first author                 | <a href="#">DOI</a>                     |
| Paper · SCIE            | Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications            | IEEE Access, vol. 9, 2021 · second author                | <a href="#">DOI</a>                     |
| Paper · SCIE            | Continual prediction of bug-fix time using deep learning-based activity stream embedding (DASENet)          | IEEE Access, vol. 8, 2020 · first author<br>33 citations | <a href="#">DOI</a>                     |
| Patent · first inventor | Graph Embedding-Based Virtual Network Mapping Method                                                        | KR 10-2629731 (2024)<br>US 12,047,250 B2 (2024)          | <a href="#">KR</a> · <a href="#">US</a> |
| Patent · first inventor | Network Monitoring System                                                                                   | KR 10-2574070 (2023)<br>US 12,556,492 B2 (2026)          | <a href="#">KR</a> · <a href="#">US</a> |
| Patent · with Samsung   | Object Storage System, Migration Control Device, and Migration Control Method                               | US 12,210,760 B2 (2025)                                  | <a href="#">US</a>                      |
| Patent · with Samsung   | Simulation Method and Modeling Method                                                                       | US 12,650,769 B2 (2026)                                  | <a href="#">USPTO</a>                   |
| Patent · with Samsung   | Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof                  | US 10,742,742 B2 (2020)                                  | <a href="#">US</a>                      |